

A 32-DIGIT PRIME x WITH $x^2 + x + 1$ SQUARE-FULL: A QUESTION OF OCHEM AND ZELINSKY ANSWERED IN THE AFFIRMATIVE

ABSTRACT. In the concluding remarks of their paper on odd perfect numbers with exactly one even exponent greater than 2, Ochem and Zelinsky ask whether there exists a prime x such that $x^2 + x + 1$ is square-full. The answer is yes:

$$x = 47116128896261596331524418282573$$

is a prime of 32 digits with $x^2 + x + 1 = 7^3 A^2$, where

$$A = 2544031832644333879196802002161,$$

and $7 \nmid A$, so every prime dividing $x^2 + x + 1$ divides it at least twice. The primality of x is certified, not merely tested: $x - 1$ is completely factored, as $2^2 \cdot 6299 \cdot 9701563 \cdot q$ with $q = 192750846182877882539$ prime, and the base 2 has full order modulo x . The witness is not ours: it is $a(27)$ of the OEIS entry A296376, whose underlying family that entry traces to a note of Majol from 1902.

1. THE QUESTION

An integer $n \geq 1$ is *square-full* (equivalently *powerful*) if $(\text{rad } n)^2 \mid n$, that is, if every prime dividing n divides it at least twice; this is the definition used in [3]. In the concluding remarks of [3], Ochem and Zelinsky pose:

Question ([3], Question 9). Does there exist a prime x such that $x^2 + x + 1$ is square-full?

They pose it because a negative answer would reduce the exponent pairs (a, b) admissible in their analysis; the affirmative answer below rules out obtaining that reduction from this question. Note that “square-full” is neither “squarefree” nor “a perfect power”.

2. THE ANSWER

Theorem 1. *Let*

$$x = 47116128896261596331524418282573.$$

Then x is prime and $x^2 + x + 1$ is square-full. Consequently the question above has the answer yes.

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Everything a reader needs is the integer x above together with the following three decimal strings. Write $n = x^2 + x + 1$ and

$$(1) \quad n = 2219929602169136991765718912586970986546248112210754339293782903,$$

$$(2) \quad A = 2544031832644333879196802002161,$$

$$(3) \quad x - 1 = 2^2 \cdot 6299 \cdot 9701563 \cdot 192750846182877882539.$$

Proof that n is square-full. Two multiplications suffice. First,

$$(4) \quad n = 7^3 A^2,$$

which is (1) against (2); and second $A \equiv 3 \pmod{7}$, so $7 \nmid A$. Now let p be any prime dividing n . If $p = 7$ then $v_7(n) = 3 + 2v_7(A) = 3 \geq 2$. If $p \neq 7$ then $p \mid A^2$ by (4), hence $p \mid A$ and $v_p(n) = 2v_p(A) \geq 2$. Every exponent in n is therefore at least 2, i.e. $(\text{rad } n)^2 \mid n$. $\square$

No factorisation of A is needed for that argument, which is why it is checkable by hand. The complete factorisation, which the argument does not use, is

$$(5) \quad \begin{aligned} n &= 7^3 \cdot 43^2 \cdot 3416683^2 \cdot 12353161^2 \cdot 1401752363863729^2, \\ A &= 43 \cdot 3416683 \cdot 12353161 \cdot 1401752363863729, \end{aligned}$$

each listed factor being prime: 43, 3416683 and 12353161 by trial division, and 1401752363863729 by the certificate of the next proof applied with base 7 to

$$1401752363863729 - 1 = 2^4 \cdot 3 \cdot 37 \cdot 107 \cdot 7376401679$$

(7376401679 prime by trial division; base 2 fails there, being a quadratic residue modulo 1401752363863729).

Proof that x is prime. This is the converse of Fermat's theorem in the form of Lucas, as in [1, Theorem 1]: if $a^{x-1} \equiv 1 \pmod{x}$ and $a^{(x-1)/q} \not\equiv 1 \pmod{x}$ for every prime $q \mid x-1$, then x is prime. Take $a = 2$. The factorisation (3) of $x-1$ is complete, and $2^{x-1} \equiv 1 \pmod{x}$, while the four quotient powers are

$$\begin{aligned} 2^{(x-1)/2} &\equiv 47116128896261596331524418282572 \pmod{x}, \\ 2^{(x-1)/6299} &\equiv 42436530233755792060895430447505 \pmod{x}, \\ 2^{(x-1)/9701563} &\equiv 21277589209484664623743014586336 \pmod{x}, \\ 2^{(x-1)/192750846182877882539} &\equiv 19695209395513058587964373225481 \pmod{x}, \end{aligned}$$

none of them 1. It remains to know that the four factors in (3) are prime. For 2, 6299 and 9701563 this is trial division. For the largest factor m we give the same certificate one level down: with

$$m = 192750846182877882539, \quad m - 1 = 2 \cdot 17^2 \cdot 27779 \cdot 12004714807399,$$

again completely factored, the base 2 satisfies $2^{m-1} \equiv 1 \pmod{m}$ and $2^{(m-1)/q} \not\equiv 1 \pmod{m}$ for each of the four primes $q \mid m-1$; and 17, 27779 and 12004714807399 are prime by trial division, the largest of them needing divisors only up to 3464782. The certificate is therefore grounded in exhaustive trial division at every leaf, and no probable-prime test enters. $\square$

Theorem 1 follows.

3. PROVENANCE OF THE WITNESS

The integer is not new. It is $a(27)$ of OEIS A296376, “Natural numbers x such that $7y^2 = x^2 + x + 1$ has a solution in natural numbers” [4], contributed by Jeffrey Shallit in December 2017, and it occurs as line 27 of the published *b*-file `b296376.txt` of that entry, computed by Colin Barker; the entry traces the family to a note of Majol from 1902 [2]. Which members of the family are square-full is immediate from the defining property rather than new: if $x = a(k)$ then $x^2 + x + 1 = 7y^2$, and such a value is square-full precisely when $7 \mid y$, since $v_7(x^2 + x + 1) = 1 + 2v_7(y)$ while every other exponent of $7y^2$ is even. No priority is claimed for the observation that this particular member is prime; our literature search was not exhaustive.

4. VERIFICATION

The file `verify.py` accompanying this note, with its recorded run in `verify.output.txt`, takes the decimal strings (1)–(3) and re-derives the mathematical assertions of Section 2: the identity $n = 7^3 A^2$ with $7 \nmid A$; the factorisation (5), with each listed prime proved prime, each exponent checked to be at least 2, and $(\text{rad } n)^2 \mid n$ verified directly; and the Lucas certificate for x together with its subsidiary certificates, every leaf grounded in exhaustive trial division. It also runs controls in both directions on its square-full decider and a census deciding every x with $1 \leq x \leq 10^8$, where the only values with $x^2 + x + 1$ square-full are the composite $x = 18$ and $x = 88916$. No minimality is claimed for the witness: nothing here excludes a prime witness with between 9 and 31 digits. Nothing bibliographic is re-derived — the statements of Section 3 are transcribed, as the program’s closing scope note, reproduced in the accompanying `REVIEW_NOTE.md`, records. The program uses exact integer arithmetic throughout, no floating-point comparison, no randomness, and the Python standard library only, and reports

VERDICT: ALL 74 CHECKS PASS

Every input it consumes is printed above. The decisive steps are elementary without being a single line of arithmetic: (4) is one multiplication and the certificate for x is five modular exponentiations, to which grounding that certificate adds the subsidiary certificate for 192750846182877882539 and the trial divisions at its leaves.

REFERENCES

- [1] J. Brillhart, D. H. Lehmer, and J. L. Selfridge, *New primality criteria and factorizations of $2^m \pm 1$* , Math. Comp. **29** (1975), 620–647. doi:10.1090/S0025-5718-1975-0384673-1.
- [2] E.-A. Majol, Note #2228, *L'Intermédiaire des Mathématiciens* **9** (1902), 183–185. Cited as the reference of [4].
- [3] P. Ochem and J. Zelinsky, *On odd perfect numbers with exactly one even exponent greater than 2*, arXiv:2607.19746v1 [math.GM], 22 July 2026. arXiv:2607.19746. The question reproduced above is Question 9, in the concluding remarks of that e-print; the numbering of a printed version, if any, may differ, which is why the question is quoted in full.
- [4] OEIS Foundation Inc., sequence A296376, *Natural numbers x such that $7y^2 = x^2 + x + 1$ has a solution in natural numbers*, contributed by J. Shallit, 11 December 2017; *b*-file `b296376.txt` (800 terms) by C. Barker. oeis.org/A296376.